PX4_DXP — Open Bug Register

3WD marking rover · branch **Upgrade_Spray @ dd166b3** (Jetson matches) · 669 tests pass · compiled 2026-07-22 from six field runs of *tes_cross_line*

A — CODE LEVEL

#	Bug	Location / parameter	Evidence	Sev
A1	**source_file never populates** → §8 ABSOLUTE ACCURACY can never run. Two different things named *source*: the stager writes a filename string, the reader expects a dict.	routes/path.py:1179 writes `"source": result["source"]` bag_autorecord.py:412 reads `source.get("filepath")`	§8 reported *unavailable* on all 6 bundles; entire absolute analysis had to be done by hand.	HIGH
A2	**must_hit lost on every non-staged load.** Only the staged path passes it, so Layer-2 vertex provenance never reaches those missions.	`mission_loading.py:100` `routes/mission.py:62` `sockets/events.py:119` `(vs routes/path.py:1279)`	mission_20260722_161014 published **must_hit = 0** on a 48-pt path.	HIGH
A3	**EKF position resets ignored.** Origin is fixed, but the vehicle estimate can jump against it mid-mission and nothing notices.	`xy_reset_counter`, `delta_xy` zero references in src/ or server/	Source-verified in ekf_helper.cpp: GNSS resets project *through* the origin and increment the counter. Residual dd166b3 does not cover.	HIGH
A4	**outcome == COMPLETE on partial runs.** Only a missing recorder_end marks INCOMPLETE; path coverage is never checked.	`bag_autorecord.py:550-560`	Two runs covered **24/64** and **76/86** waypoints — both reported COMPLETE.	HIGH
A5	**No /spray/session_config subscription.** No mode can reach the spray node; dash and point remain schema-only.	`spray_controller_node.py:444` `(node's own comment)`	Blocks Spray V2 Phase C and D entirely.	HIGH
A6	**Spray plan Rev 3 stale vs shipped code.** §5/§7.2/§7.3 describe a speed gate 6523a84 removed, and deny an RPP coupling that now exists.	`SPRAY_CONTROLLER_V2_PLAN.md` `vs` `spray_controller_node.py:877`	Implementing Phase C against it repeats the 16480d9 false-fix pattern.	HIGH
A7	**Cross-session origin repeatability.** Origin re-derived at each EKF restart from a fresh first fix; needs SET_GPS_GLOBAL_ORIGIN pinning at boot.	`not implemented anywhere`	One degE7 quantisation step (≤1.11 cm) of offset per session. Within a session, identical.	MED
A8	**CPU contention — executor mismatch.** MultiThreadedExecutor with MutuallyExclusiveCallbackGroup gives zero parallelism.	`ros_node.py executor setup`	50–65% of one core. Arm/disarm safety blocks a naive swap.	MED

B — CALIBRATION / HARDWARE (QGC parameters)

#	Bug	Location / parameter	Evidence	Sev
B1	**Dual-antenna heading bias — largest open error.** A constant heading error makes pure pursuit settle on a line parallel to the path.	`EKF2_GPS_YAW_OFF` default 0.0, 0–360 deg (QGC)	**5 of 6 runs drove left**; B spread 3.74 cm, means –0.47 to –2.00 cm. offset ≈ lookahead × sin(err): 2 cm needs only **~2.2°** at 0.52 m lookahead.	HIGH
B2	**Differential wheel asymmetry.** One counts-per-rev value for two wheels that may not match; same one-sided signature as B1.	`RBCLW_COUNTS_REV` `RD_WHEEL_TRACK` (default 0)	Indistinguishable from B1 in the data. Test B1 first — cheaper.	MED
B3	**GNSS antenna lever arm.** Antenna off the vehicle centreline gives a fixed lateral offset; roll amplifies it.	`EKF2_GPS_POS_X/Y/Z`	Reference rig: 1.934 m pole, 1.7° tilt = 5.7 cm lateral if uncompensated.	MED
B4	**Nozzle offset uncalibrated.** Every measurement to date describes where the *antenna* went, not where paint went.	`nozzle_forward_offset_m` `nozzle_lateral_offset_m` declared, unmeasured	Blocks error budget 4. TinySurveyor logs antenna and tool coordinates separately for this reason.	HIGH
B5	**Physical re-survey never done.** No log can answer where the paint landed.	`process gap — Emlid RS3`	Survey CSV shows `Samples=1`, 1.7 cm single-epoch RMS. Use averaging when re-surveying.	HIGH

C — SHIPPED BUT UNVERIFIED (code-proven, no field evidence)

#	Bug	Location / parameter	Evidence	Sev
C1	Placement determinism (dd166b3)	`origin acquired live; unit-tested`	No two-load field comparison. The core claim is untested on hardware.	HIGH
C2	Survey CSV ingest (211244b)	`unit tests only`	Never run in the field — all missions to date were DXF.	MED
C3	POINT-layer control points	`synthetic road case only`	Never exercised on a dense real drawing.	MED
C4	§8 absolute accuracy	`blocked by A1`	Has never auto-run on a real bundle.	HIGH
C5	segment→smooth ~5 cm seam	`visual only`	May already be fixed by 64c12ff. Check /rpp/debug[0] vs [40] at the profile flip.	MED
C6	Spray pivot-state gate (6523a84)	`replay-verified 157→53`	Field-unverified.	MED
C7	Endpoint under-run	`measured, no fix attempted`	Rover rests 0.9–2.0 cm short every run; overshoot never above +0.4 cm.	MED
C8	Per-run wander	`measured, untouched`	Residual RMS 1.1–1.8 cm between runs after removing the mean offset.	MED

Recommended order. 1. B1 heading test — largest error, no driving needed, one QGC parameter. 2. A1, A2, A4 — small, and all three currently corrupt the measurements you collect. 3. C1 two-load determinism check — one minute, closes dd166b3. 4. B4 + B5 — the only route to knowing where paint lands. 5. A6 → A5 — unblocks spray modes.

Field-proven fixes this cycle: vertex deletion (64c12ff — 64→4 conditioned points, was 64→2) and georef north-scale (e483d53 — 2.3255 m vs a 2.3255 m WGS84 geodesic, residual 0.000 mm). Tracking is sound at 0.33–0.88 cm per surveyed vertex with extensions; the remaining error is offset, not tracking.